Overview, Design and Details (ODD) (Grimm et al., 2017) Protocol for: ***Controlling the misinformation diffusion in social media by the effect of different classes of agents.*** (Khodabandeh Yalabadi et al., 2023) (Table 1)

Table 1: Overview, Design and Details (ODD) of the model

Section	Description
1. Purpose overview	An agent-based framework to simulate the diffusion process of a piece of misinformation according to the SBFC model in which the fake news and its debunking compete in a social network. Considering new classes of agents, this model is closer to reality and proposed different strategies how to mitigate and control misinformation.
2. Entities, state variables and scales	The entities are user accounts in social media that can spread their belief and information. Each agent has three states (Susceptible, Believer, and Fact-checker). Agents have characteristics of the probabilities of verifying the hoax and forgetting their state. Additionally, there are four classes of agents (scholar, influencer, bots (super-spreaders), and normal people) with specific probabilities, community sizes, and behavior. The input variables are the cluster of scholars, spreading rate, hoax credibility, % of initial believers, % of each type of bot (believer, or fact-checker) in the network, and probabilities of verifying and forgetting for each class. All the scales are normal scales (0-1) and percentage scales.
3. Process overview and scheduling	First, the inputs will be modified and then we considered using Facebook (social circles) dataset (Leskovec et. al., 2012) instead of generating a random network in each run. This network has 4039 nodes, 88234 edges, and the average degree of nodes is 43.7. With this network, we can have fixed clusters, and node degrees which are helpful for assigning influencers and bots. We need clusters because we want to assign the scholar class to one of these clusters. For clustering, the asynchronous fluid community's algorithm (Pares et. al., 2018) was used to generate eight clusters. Different classes will be assigned to each agent. The top 1% of agents in terms of node degree will be influencers and bots (super-spreaders) will be randomly assigned to the top 5% high node degree agents (excluding influencers).
4. Design concepts	The dynamic of this model and the way of transitioning between states comes from previous works on the SBFC model (Tambuscio et. al., 2015, Sulis, and Tambuscio, 2020). Adding two critical classes of influencers and bots are inspired by Beskow and Carley's (2019) work.
	This SBFC describes misinformation diffusion as a competition between a piece of fake news and its debunking. Consider that we have a population of N agents, and each agent i at each time t has a state of $S_i(t)$ that can be one of the following: • Susceptible (S) is an agent who will change its state based on the news. • Believer (B), an agent who believes to the hoax. • Factchecker (FC), an agent who has verified the news or directly knows that it is a hoax.

As it was mentioned, misinformation and fact-checking are competing actors and the main factors are the credibility of the hoax, the spreading rate, the belief of neighbors, probability of verifying or forgetting the news.

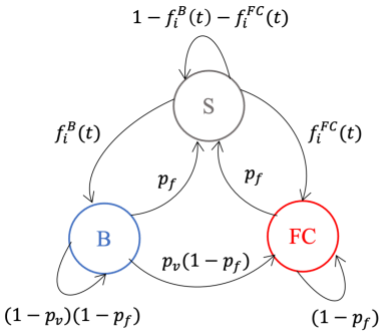


Figure 1: dynamics of SBFC model (Tambuscio (2015))

The dynamics are shown in Fig. 1. There are two spreading transitions ($f_i^B(t), f_i^{FC}(t)$) that make a susceptible (S) agent become a believer (B) or a fact checker (FC). These transitions are defined based on hoax credibility (α), spreading rate (β), and belief of neighbors ($n_i^B(t)$ believers, $n_i^{FC}(t)$ fact checkers):

$$f_i^B(t) = \beta \frac{n_i^B(t)(1+\alpha)}{n_i^B(t)(1+\alpha) + n_i^{FC}(t)(1-\alpha)}$$

$$f_i^{FC}(t) = \beta \frac{n_i^{FC}(t)(1-\alpha)}{n_i^B(t)(1+\alpha) + n_i^{FC}(t)(1-\alpha)}$$

Where:

$$n_i^B(t) = n_{Sc_i}^B(t) + n_{I_i}^B(t) + n_{N_i}^B(t) + n_{Bot_i}^B(t)$$

$$n_i^F(t) = n_{Sc_i}^F(t) + n_{I_i}^F(t) + n_{N_i}^F(t) + n_{Bot_i}^F(t)$$

We have four different classes of agents in the network with their specific characteristics:

- The scholars ("Sc") who have a clearer understanding of information, can verify the news better, remember the news credibility longer, and their community is concentrated in a cluster, so they get more effect from each other as one cluster.
- The influencers ("I") who have more influence because of their big neighborhood size, stay in their state (believer, fact-checker) longer, and could have a more verifying probability.
- Bots or Super-spreaders ("Bot") who do not change their state and do not forget. They just act as spreaders in the network. They can be believer bots that are programmed to spread the hoax, or they can be fact-checker bots that are programmed to mitigate the misinformation diffusion.
- The normal ("N") agents who forget their state more frequently and have less verifying probability.

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